# PU WANG

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## EDUCATION

**College of Computer Science and Technology, Zhejiang University** — **Hangzhou, China**
B.E. in Artificial Intelligence (Turing Class) — *Sep 2023 – Jun 2027 (expected)*
Honors degree from Chu Kochen Honors College
Related courses: *Introduction to Theoretical Computer Science, Introduction to Natural Language Processing, Machine Learning, Mathematical Analysis I/II*

## RESEARCH INTERESTS

My research interests lie in the theory and algorithms of **sequential decision making**, with a growing focus on **large language models** and **language agents**. I am interested in how agents acquire, through interaction, the knowledge they need to act well, and how that knowledge can be learned efficiently and reused across tasks and environments. More broadly, while algorithms and theory remain at the core of my work, I aim to bring principled formulations from bandits and reinforcement learning to problems related to decision making in language models and language agents.

## PUBLICATIONS

**Tree-Guided Identify-Then-Exploit: A Unified Framework of Best Arm Identification and Regret Minimization for Dueling Bandits.**
***Pu Wang***, *Yao-Xiang Ding*
***NeurIPS 2026***: *The Fortieth Annual Conference on Neural Information Processing Systems*

## RESEARCH EXPERIENCE

**Research Intern at State Key Lab of CAD&CG** — Mar 2025 – Present
*Advisor: Prof. Yao-Xiang Ding* — Zhejiang University

- **TG-ITE: A Unified Framework for Dueling Bandits** — *Mar 2025 – May 2026*
  Developed the first unified framework for $N$-armed dueling bandits that jointly handles best arm identification (BAI), weak regret, and strong regret under only the Condorcet-winner assumption. It achieves $O(N)$ sample complexity for BAI and $O(N)$ weak regret, and **removes the $O(\log N)$ suboptimality gap** of the existing approach when optimizing both jointly.
- **Imitation Learning from Partially Observable Experts** — *Jun 2025 – Dec 2025*
  Studied imitation from multiple experts that each observe only a masked part of the state. Proved a **negative identifiability result**: mask coverage alone cannot recover the causal mask or the optimal policy. Proposed **POIL**, a GAIL-based method with active querying, with preliminary MuJoCo experiments.

## PROJECTS

**Needle: A Compiler from C to RISC-V** — Mar 2025 – Jun 2025
*Course Project for Principles of Compilers* — Zhejiang University

- **Frontend & IR:** Implemented an end-to-end compiler for a C subset using *Flex* and *Bison*, featuring AST construction, semantic analysis, and Intermediate Representation (IR) generation.
- **Code Generation:** Developed a RISC-V code generator handling calling conventions and control flow, verifying correctness through simulation tools.

**Linear Algebra: Left Undone**
*Open-Source Supplementary Lecture Notes* — Chu Kochen Honors College, Zhejiang University

- **Overview:** Open-source supplementary linear algebra lecture notes of Chu Kochen Honors College, Zhejiang University, with 700+ stars on GitHub.
- **Contribution:** Authored Chapter 24, *Linear Algebra and Calculus*.

## ACHIEVEMENTS

Zhejiang University Scholarship – Second Prize *2024, 2025*
Second Class Scholarship for Elite Students in Basic Sciences *2025*

## SKILLS

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|---|---|
| **Programming Languages** | Python, C/C++ |
| **Machine Learning Tools** | PyTorch, einx/einops, Gymnasium, Stable-Baselines3 |
| **Systems & Infra** | Linux, Git, Shell Scripting, Docker, Slurm |
| **Writing & Typesetting** | LaTeX, Typst, TikZ |

I maintain an open technical notebook (400,000+ words) on computer science, mathematics, and AI, as a long-term knowledge-building practice.